E. Coli Klebsiella Proteus and Enterobacter spp.

GUS-3rd year

LG2

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Learning outcomes

- Recognize the main clinical syndromes caused by these organisms.
- Link virulence factors to urinary tract disease and complications
- Interpret urine cultures with the clinical presentation.
- Explain treatment selection, resistance and source control.

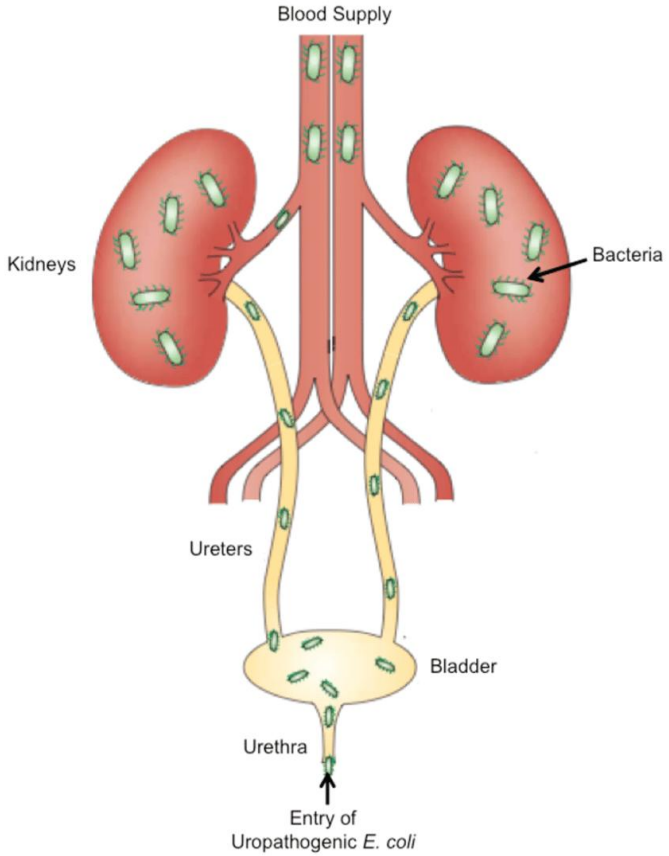
Shared microbiological features



- All belong to the order **Enterobacterales**.
- **Gram-negative**, non-spore-forming rods.
- Facultative anaerobes that ferment glucose.
- The four groups are usually **oxidase negative** and reduce nitrate.
- Gut carriage can provide a **reservoir** for urinary infection.

Ascending urinary tract infection

- Gut organisms reach the **periurethral** area.
- They **ascend** the urethra and colonize the bladder.
- **Adhesion** helps bacteria resist removal by urine flow.
- Further ascent can involve the kidneys and bloodstream.



Clinical syndromes

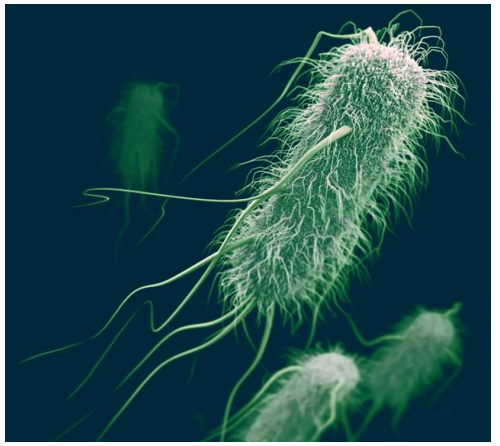
Syndrome	Typical findings
Cystitis	Dysuria, frequency, urgency, suprapubic discomfort
Pyelonephritis	Fever, flank pain, costovertebral angle tenderness
Acute bacterial prostatitis	Fever, pelvic or perineal pain, urinary symptoms
Urosepsis	Urinary infection with life-threatening organ dysfunction

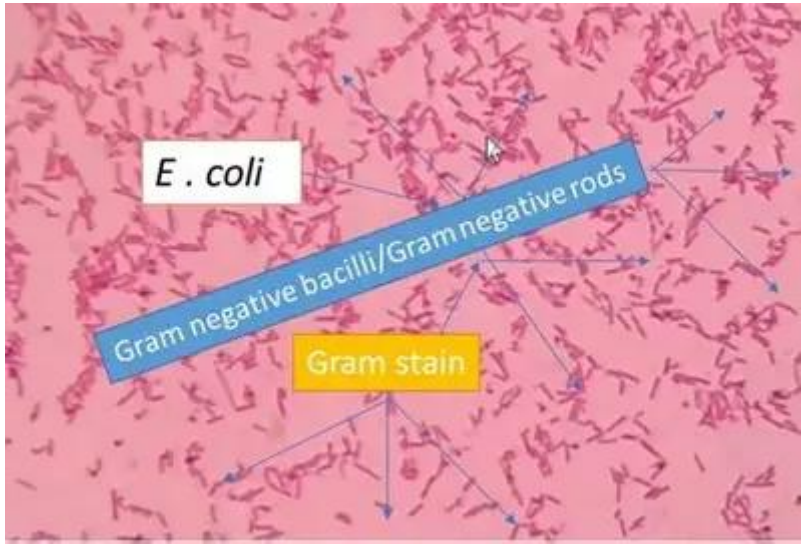
Factors that complicate the clinical course

- Obstruction, stones, incomplete bladder emptying.
- Urinary catheters and recent instrumentation.
- Pregnancy, impaired immunity or renal impairment.
- Recent antibiotics or a previous resistant urine isolate.

Escherichia coli

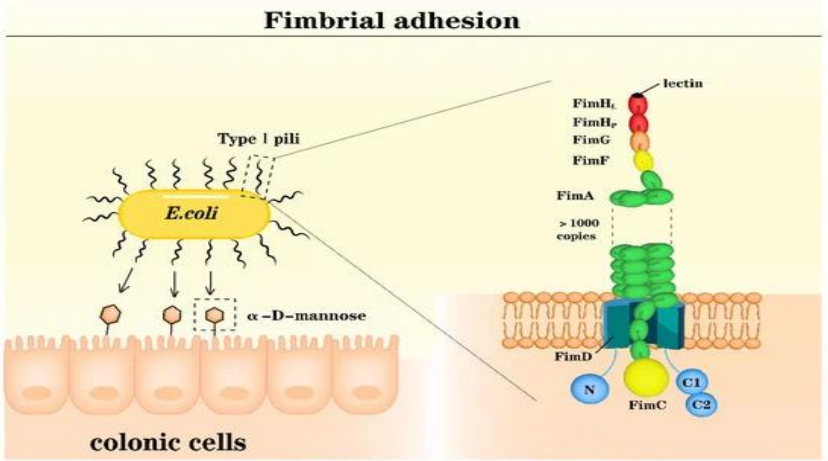
- A major member of the normal intestinal microbiota.
- Usually **motile** and **lactose** fermenting.
- Typically **indole** positive and urease negative.
- Uropathogenic strains are termed **UPEC**.

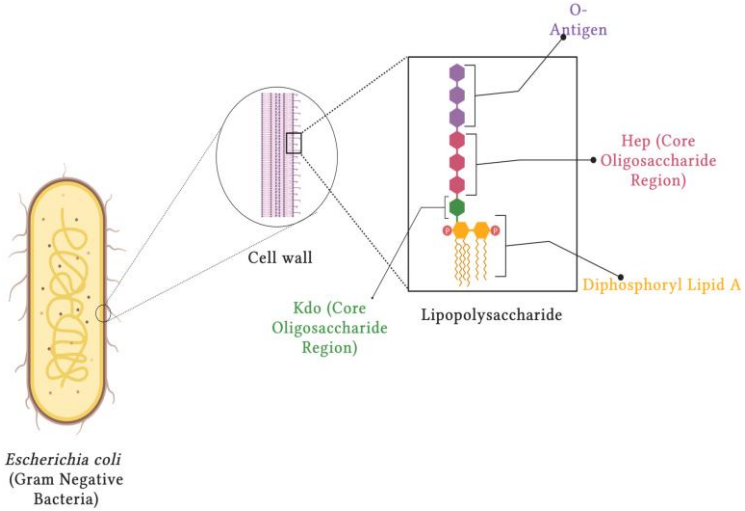




Virulence Factors

- **Fimbriae**: Bladder attachment and invasion
- **Capsule** and surface structures: Support survival against host defenses
- **Hemolysin** and other toxins: Can injure host cells
- **Lipopolysaccharide**: Triggers inflammation during local or invasive disease





Clinical disease caused by *E. coli*

- Acute **cystitis** is the common urinary presentation.
- Ascending infection can cause **pyelonephritis**.
- **Prostatic** infection may occur, especially with fever and pelvic pain.
- Urinary infection can progress to **bacteremia** and sepsis.

Clinical presentation

- **Cystitis:**

Dysuria with frequency and urgency, Suprapubic discomfort

- **Pyelonephritis:**

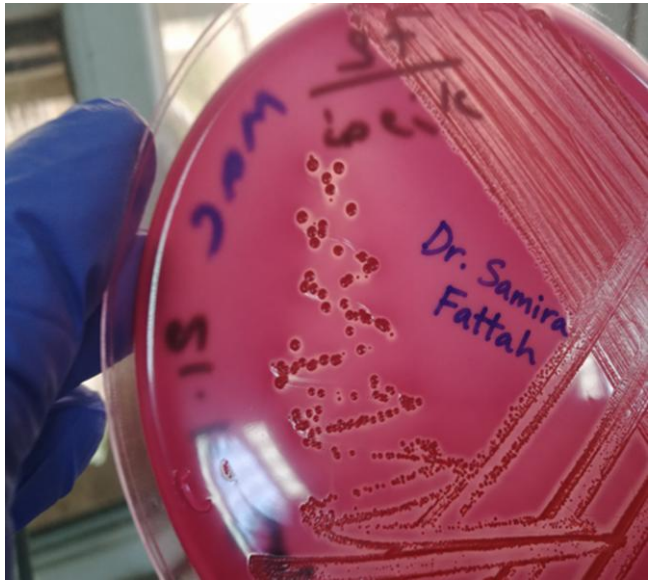
Fever, flank pain, nausea or vomiting are typical, Complications include bacteremia, renal abscess and acute kidney injury. Obstruction with infection requires urgent urological assessment.

- **Prostatitis:**

Fever, perineal pain or urinary retention. Treatment requires appropriate tissue penetration and a longer course.

Laboratory diagnosis of *E. coli*

- MacConkey agar usually shows pink **lactose-fermenting** colonies.
- **Indole** is usually **positive** and citrate usually negative.
- An identification automated system confirms identity.
- AST is **needed** when culture is clinically indicated.
- Colony appearance **cannot** establish the UPEC pathotype.



Klebsiella

- *K. pneumoniae* is the main species in clinical practice.
- **Encapsulated** Gram-negative rods.
- Classically **nonmotile** and **lactose fermenting**.
- Colonies often have a **mucoid** appearance.
- Colonizes the **intestine** and causes opportunistic infections.
- Some isolates form a viscous string when lifted with a loop.

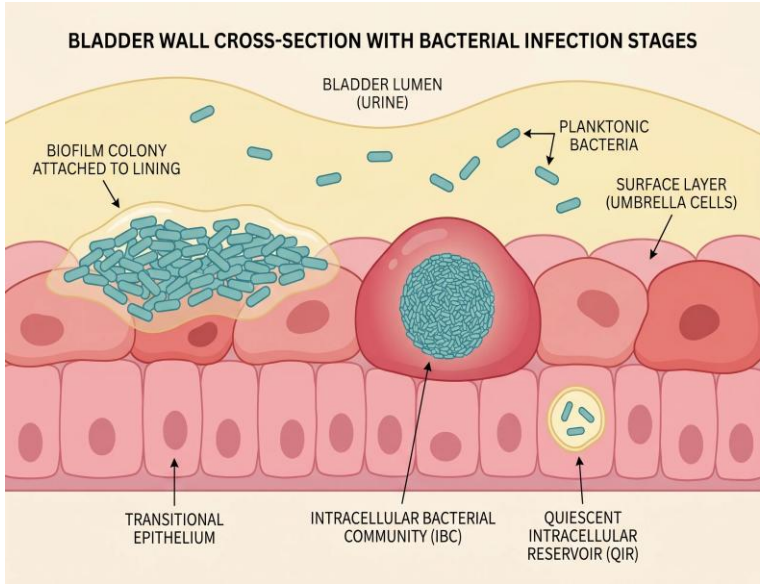






Virulence factors

- **Polysaccharide capsule**: Helps resist phagocytosis and host killing
- **Fimbriae**: Support epithelial or device attachment
- **Biofilm formation**: Supports persistence on urinary devices.



Klebsiella urinary disease

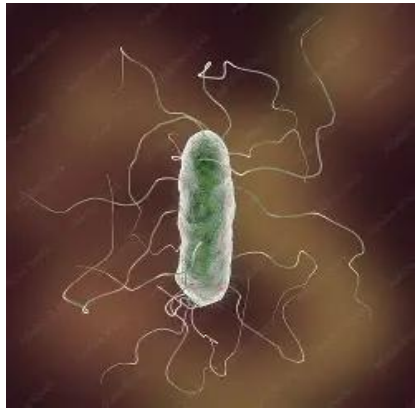
- Causes cystitis, pyelonephritis and urinary-source bacteremia.
- Cause UTI after catheterization or urinary instrumentation.
- Hospital **exposure** and **recent** antibiotics raise resistance concerns.
- Symptoms overlap with *E. coli* infection.
- *K. pneumoniae* has **expected resistance** to ampicillin.
- Extended-spectrum β-lactamases (**ESBLs**) and **carbapenemases** may restrict treatment.

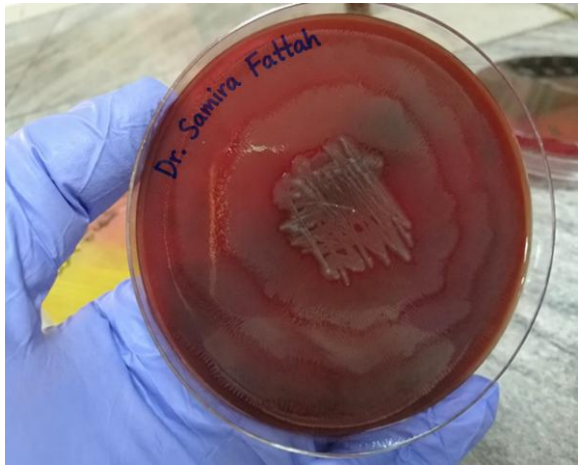
E. coli and *K. pneumoniae*

Typical feature	***E. coli***	***K. pneumoniae***
Lactose sugar fermentation	Usually positive	Usually positive
Motility	Usually positive	Negative
Indole	Usually positive	Usually negative
Urease	Usually negative	Usually positive

Proteus spp.

- *P.* ***mirabilis*** is the principal urinary species.
- *P.* ***vulgaris*** also causes opportunistic infection.
- **Highly motile** rods, typically **non-lactose** fermenting.
- **Swarming** can produce concentric growth on blood agar.
- Strong urease activity is the key clinical association.



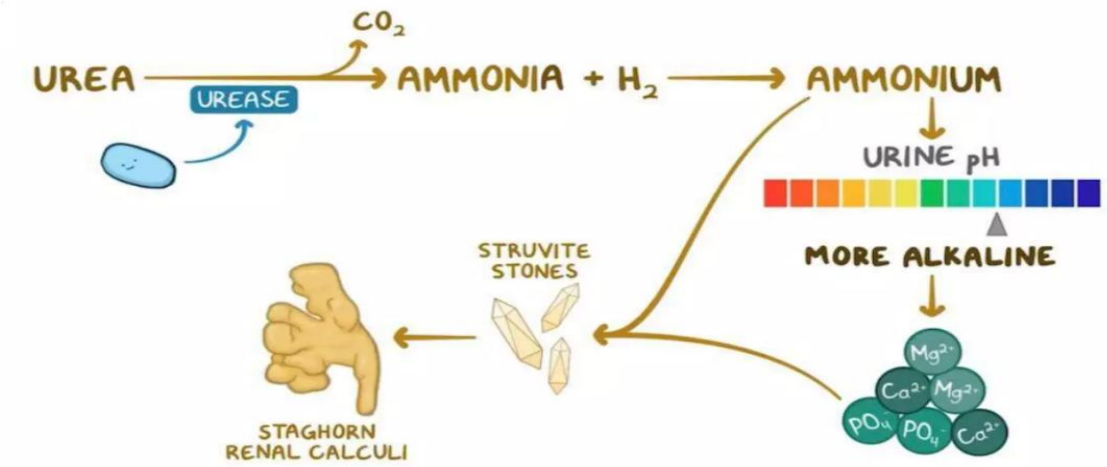




colorless colonies with slight swarming

Proteus urease and alkaline urine

- Urea hydrolysis: Urea + water forms ammonia and carbon dioxide
- Ammonia production: Urinary pH rises↑
- Alkaline conditions: Struvite and carbonate apatite can precipitate
- Crystalline deposits: Stones and catheter encrustation can develop.

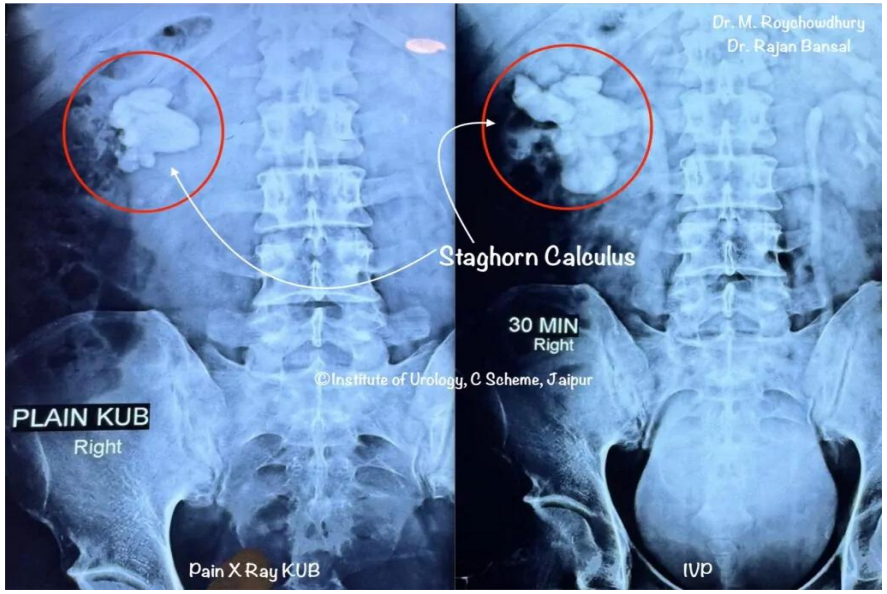


Infection stones and obstruction

- Stones can **protect** bacteria and contribute to recurrence.
- Struvite stones may become large branching stones or form a **staghorn calculus**.
- Obstruction can impair drainage and damage the kidney.



STAGHORN CALCULUS



Proteus and urinary catheters biofilm

- **Adhesion** and biofilm support growth on catheter surfaces.
- **Urease** can cause crystalline biofilm and encrustation.
- A blocked catheter can cause **retention** and impaired **drainage**.
- Catheter problems and infection require clinical assessment.
- Long-term catheter bacteriuria alone is not an indication for antibiotics.

Fig 2. **Catheter encrustation**



Fig 2a. Encrustation can form on the outside of the catheter



Fig 2b. Encrustation can form on the inside wall of the catheter

Laboratory Diagnosis

Test or feature	***P. mirabilis***	***P. vulgaris***
Lactose fermentation	Negative	Negative
Urease	Positive	Positive
Motility / swarming	Positive	Positive
Indole	**Negative**	**Positive**
Hydrogen sulfide	Usually positive	Usually positive

Management of Proteus UTI

- Choose an agent supported by AST and the infection site.
- Nitrofurantoin is an **unsuitable** choice for Proteus. Why?
- Persistent or recurrent infection **raises concern** for stones.
- Restore urinary drainage when obstruction is present.
- ESBL or other acquired resistance can occur.

Enterobacter spp.

- The ***E. cloacae* complex** is a group of several related species (*E. cloacae*, *E. hormaechei*, *E. kobei*, *E. roggenkampii*, *and E. asburiae*) .
- **Routine** testing could not tell them apart.
- Newer identification methods are closing that gap: multiplex PCR, MALDI-TOF mass spectrometry.
- Usually **motile**, glucose fermenting and often **lactose-fermenting**.
- Colonizes the **gut** and can persist in healthcare environments.
- Causes catheter-associated UTI, bacteremia and other opportunistic infections.



Enterobacter: clinical treatment implications

- *E. cloacae* complex has a meaningful risk of **AmpC overproduction**.
- Ceftriaxone is generally **avoided** for invasive infection.
- Cefepime (4th gen) may be appropriate when active and correctly dosed.
- A **carbapenem** may be needed for severe disease or additional resistance.
- Treatment decisions include severity and source control.

AmpCs can hydrolyze third-generation cephalosporins and piperacillin/tazobactam

Comparative laboratory identification

Typical feature	*E. coli*	*K. pneumoniae*	*P. mirabilis*	*E. cloacae* complex
Lactose	+	+	–	Usually +
Motility	+	–	+	+
Indole	+	–	–	Usually –
Urease	–	+	Strong +	Variable
Citrate	–	+	+	+

Useful clue *	Common UTI	Mucoid	Swarming	AmpC risk

Treatment

- **Nitrofurantoin**: Bladder infection only. Unsuitable for Proteus.
- **Fosfomycin**: An option for selected *E. coli* cystitis.
- **ESBL-producing**: TMP-SMX or a fluoroquinolone may be suitable in complicated UTI.
- **Inducible AmpC** risk: Resistance may emerge with some cephalosporins.
- **Carbapenem resistance**: AST guide specialist treatment

Asymptomatic bacteriuria

- Bacteriuria occurs without symptoms attributable to UTI.
- Most nonpregnant adults do not benefit from antibiotic treatment.
- This includes many older adults, people with diabetes and catheter users.
- Key exceptions: pregnancy and urological procedures with mucosal trauma.

Pregnancy and other special situations

- Pregnancy **requires** screening and treatment of asymptomatic bacteriuria.
- Choose antibiotics according to **gestation**, **susceptibility** and **safety**.
- Pyelonephritis in pregnancy needs prompt assessment.
- **Children**, **severe renal impairment** and **immunocompromise** need tailored care.

Empirical antibiotic selection

- **Identify the site**: bladder, kidney, prostate or bloodstream.
- **Assess severity**: including sepsis and urinary obstruction.
- **Review earlier cultures, recent antibiotics** and the **local antibiogram**.
- Reassess after culture results and narrow treatment when appropriate.

Case1 : recurrent infection and alkaline urine

- A 68-year-old man with a long-term catheter develops fever and flank pain.
- His catheter drains poorly. Urine pH is 8.5.
- Culture grows a swarming, urease-positive, indole-negative organism.
- Which organism and complications are most likely?

Case 2: a positive culture without symptoms

- A 74-year-old woman with diabetes has a routine urine culture.
- *K. pneumoniae* grows at 10^5 CFU/mL with pyuria.
- She has no dysuria, fever, flank pain or new urinary symptoms.
- Does she need antibiotics? What circumstances would change the answer?

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